

Advanced Coordinate Geometry

ANSWER KEY



Distance, midpoint, and slope combined

- 1 Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB .

Midpoint of AB : $\left(\frac{-2+4}{2}, \frac{1+1}{2}\right) = (1, 1)$. Length of median from $C(1, 7)$ to $(1, 1)$:
 $\sqrt{(1-1)^2 + (7-1)^2} = \sqrt{0+36} = 6$.

Equations of circles from given conditions

- 2 Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.

Center (midpoint): $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$. Radius (half the diameter length): diameter
 $= \sqrt{(8-2)^2 + (11-3)^2} = \sqrt{36+64} = \sqrt{100} = 10$, so radius $= 5$. Equation:
 $(x-5)^2 + (y-7)^2 = 25$.

Distance from a point to a line

- 3 Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- $$d = \frac{|3(3) - 4(4) + 5|}{\sqrt{9+16}} = \frac{|9-16+5|}{5} = \frac{|-2|}{5} = \frac{2}{5}.$$
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Locus problems

- 4 Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

The locus of points equidistant from two fixed points is the perpendicular bisector of the segment joining them. Since both points lie on the x -axis symmetric about the origin, the perpendicular bisector is the vertical line $x = 0$ (the y -axis).

Rotations and general transformations

- 5 A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \rightarrow (-y, x)$.

Applying the rule: $(4, 0) \rightarrow (-0, 4) = (0, 4)$.

Dilations and scale factor

- 6 A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.

Multiply each coordinate by 3: $(3, 6)$, $(9, 6)$, $(3, 18)$.

Finding the equation of a parabola from three points

- 7 A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a , b , and c .

From $(0, 3)$: $c = 3$. From $(1, 6)$: $a + b + c = 6$, so $a + b = 3$. From $(-1, 2)$: $a - b + c = 2$, so $a - b = -1$. Adding $a + b = 3$ and $a - b = -1$: $2a = 2$, so $a = 1$, then $b = 2$. Equation: $y = x^2 + 2x + 3$.

A multi-part triangle-in-the-plane problem

- 8 This question has two parts. Triangle POR has $P(0, 0)$, $O(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

(a) $\text{Area} = \frac{1}{2}|0(0 - 6) + 6(6 - 0) + 3(0 - 0)| = \frac{1}{2}|0 + 36 + 0| = 18$. (b) Using base $PQ = 6$ (along the x -axis) and height equal to R 's y -coordinate, 6: $\text{Area} = \frac{1}{2}(6)(6) = 18 \checkmark$, confirming both methods agree.

Finding the equation of a line given a point and a perpendicular condition

- 9 Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

Midpoint: $\left(\frac{2+8}{2}, \frac{3+7}{2}\right) = (5, 5)$. Slope of the segment: $\frac{7-3}{8-2} = \frac{4}{6} = \frac{2}{3}$. Perpendicular slope: $-\frac{3}{2}$.
Equation: $y - 5 = -\frac{3}{2}(x - 5)$, so $y = -\frac{3}{2}x + \frac{15}{2} + 5 = -\frac{3}{2}x + \frac{25}{2}$.

Translating and reflecting equations of curves

- 10 The graph of $y = x^2$ is reflected over the x -axis, then shifted 3 units up. Write the equation of the resulting graph.

Reflecting over the x -axis: $y = -x^2$. Shifting up 3: $y = -x^2 + 3$.

Finding vertices of a polygon given transformations

- 11 A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

Add $(3, -2)$ to each vertex: $(3, -2)$, $(7, -2)$, $(7, 2)$, $(3, 2)$.

Finding the intersection of a line and a parabola

- 12 Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

Set equal: $x^2 - 2x - 3 = x - 3$, so $x^2 - 3x = 0$, giving $x(x - 3) = 0$, so $x = 0$ or $x = 3$. Corresponding y -values: for $x = 0$, $y = -3$; for $x = 3$, $y = 0$. Intersection points: $(0, -3)$ and $(3, 0)$.

A multi-part quadrilateral classification problem

- 13 This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY , and of WZ and XY . (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

(a) $WX = \sqrt{(6-0)^2 + 0^2} = 6$. $XY = \sqrt{(8-6)^2 + 4^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$.
 $YZ = \sqrt{(2-8)^2 + 0^2} = 6$. $ZW = \sqrt{(0-2)^2 + (-4)^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$. (b) Slope of WX : $\frac{0-0}{6-0} = 0$. Slope of ZY : $\frac{4-4}{8-2} = 0$ (parallel, since equal slopes). Slope of WZ : $\frac{4-0}{2-0} = 2$. Slope of XY : $\frac{4-0}{8-6} = 2$ (also parallel, equal slopes). (c) Since both pairs of opposite sides are parallel ($WX \parallel ZY$ and $WZ \parallel XY$) and opposite sides are equal in length ($WX = YZ = 6$, $XY = ZW = 2\sqrt{5}$), the quadrilateral is a parallelogram.

Finding the equation of a tangent line to a circle

- 14 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

The radius to $(3, 4)$ has slope $\frac{4-0}{3-0} = \frac{4}{3}$. The tangent line is perpendicular, so its slope is $-\frac{3}{4}$. Equation: $y - 4 = -\frac{3}{4}(x - 3)$, so $y = -\frac{3}{4}x + \frac{3}{4} + 4 = -\frac{3}{4}x + \frac{25}{4}$.

Finding the vertex and axis of symmetry of a parabola from three points

- 15 A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.

Since the parabola has zeros at $x = -1$ and $x = 3$, the axis of symmetry lies midway between them: $x = \frac{-1+3}{2} = 1$. Since $(1, -8)$ lies exactly on the axis of symmetry $x = 1$, this point IS the vertex: $(1, -8)$.

Multiple choice — easy

- 16 Find the slope of the line through $(4, -5)$ and $(8, -2)$.

- a. $-\frac{3}{4}$
- b. $\frac{3}{4}$ ✓
- c. $-\frac{4}{3}$
- d. $\frac{4}{3}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 + 5}{8 - 4} = \frac{3}{4}.$$

- 17 Find the slope of the line through $(3, -6)$ and $(2, 2)$.

- a. $\frac{1}{8}$
- b. 8
- c. $-\frac{1}{8}$
- d. -8 ✓

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 + 6}{2 - 3} = -8.$$

- 18 Find the slope of the line through $(-6, -5)$ and $(-7, 6)$.

- a. -11 ✓
- b. $-\frac{1}{11}$
- c. 11
- d. $\frac{1}{11}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 + 5}{-7 + 6} = -11.$$

19 Find the slope of the line through $(-3, 8)$ and $(-9, 5)$.

a. 2

b. -2

c. $\frac{1}{2}$ ✓

d. $-\frac{1}{2}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 8}{-9 + 3} = \frac{1}{2}.$$

20 Find the slope of the line through $(-4, 6)$ and $(5, 9)$.

a. -3

b. $\frac{1}{3}$ ✓

c. $-\frac{1}{3}$

d. 3

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{9 - 6}{5 + 4} = \frac{1}{3}.$$

21 Find the slope of the line through $(-6, 6)$ and $(-5, 6)$.

a. 0 ✓

b. 1

c. 2

d. 3

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 6}{-5 + 6} = 0.$$

22 Find the slope of the line through $(-9, -5)$ and $(-6, -8)$.

a. 1

b. 2

c. -1 ✓

d. 0

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-8 + 5}{-6 + 9} = -1.$$

- 23 Find the slope of the line through $(-1, -9)$ and $(4, 0)$.

- a. $-\frac{9}{5}$
- b. $\frac{5}{9}$
- c. $\frac{9}{5}$ ✓
- d. $-\frac{5}{9}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 + 9}{4 + 1} = \frac{9}{5}.$$

- 24 Find the slope of the line through $(9, 0)$ and $(-6, 4)$.

- a. $-\frac{4}{15}$ ✓
- b. $\frac{15}{4}$
- c. $-\frac{15}{4}$
- d. $\frac{4}{15}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4}{-6 - 9} = -\frac{4}{15}.$$

- 25 Find the slope of the line through $(-5, -1)$ and $(-4, 8)$.

- a. -9
- b. 9 ✓
- c. $\frac{1}{9}$
- d. $-\frac{1}{9}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{8 + 1}{-4 + 5} = 9.$$

- 26 Find the slope of the line through $(9, -1)$ and $(-2, 9)$.

- a. $\frac{10}{11}$
- b. $-\frac{10}{11}$ ✓
- c. $-\frac{11}{10}$
- d. $\frac{11}{10}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{9 + 1}{-2 - 9} = -\frac{10}{11}.$$

- 27 Find the slope of the line through $(-3, 7)$ and $(7, -4)$.

- a. $-\frac{11}{10}$ ✓
- b. $\frac{10}{11}$
- c. $\frac{11}{10}$
- d. $-\frac{10}{11}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 7}{7 + 3} = -\frac{11}{10}.$$

- 28 Find the slope of the line through $(-8, -1)$ and $(-1, 5)$.

- a. $\frac{7}{6}$
- b. $-\frac{6}{7}$
- c. $-\frac{7}{6}$
- d. $\frac{6}{7}$ ✓

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 + 1}{-1 + 8} = \frac{6}{7}.$$

- 29 Find the slope of the line through $(8, -9)$ and $(6, 4)$.

- a. $\frac{2}{13}$
- b. $\frac{13}{2}$
- c. $-\frac{2}{13}$
- d. $-\frac{13}{2}$ ✓

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 + 9}{6 - 8} = -\frac{13}{2}.$$

30 Find the slope of the line through $(-9, -3)$ and $(-7, 1)$.

a. 2 ✓

b. -2

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 + 3}{-7 + 9} = 2.$$

Multiple choice — medium

31 Find the distance from the point $(9, -1)$ to the line $20x - 21y - 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{248}{29}$

b. $\frac{219}{29}$

c. -190

d. $\frac{190}{29}$ ✓

$$d = \frac{|20(9) - 21(-1) - 11|}{\sqrt{400 + 441}} = \frac{|190|}{29} = \frac{190}{29}.$$

32 Find the distance from the point $(1, -1)$ to the line $3x - 4y + 6 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{18}{5}$

b. $\frac{13}{5}$ ✓

c. $\frac{23}{5}$

d. -13

$$d = \frac{|3(1) - 4(-1) + 6|}{\sqrt{9 + 16}} = \frac{|13|}{5} = \frac{13}{5}.$$

- 33 Find the distance from the point $(8, -6)$ to the line $5x + 12y - 4 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $-\frac{36}{17}$

b. $\frac{36}{17}$

c. $\frac{36}{13}$ ✓

d. $-\frac{36}{13}$

$$d = \frac{|5(8) + 12(-6) - 4|}{\sqrt{25 + 144}} = \frac{|-36|}{13} = \frac{36}{13}.$$

- 34 Find the distance from the point $(0, 3)$ to the line $8x + 15y - 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{70}{17}$

b. $\frac{36}{17}$ ✓

c. $\frac{53}{17}$

d. $\frac{36}{23}$

$$d = \frac{|8(0) + 15(3) - 9|}{\sqrt{64 + 225}} = \frac{|36|}{17} = \frac{36}{17}.$$

- 35 Find the distance from the point $(9, 7)$ to the line $5x + 12y + 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{166}{13}$

b. $\frac{140}{17}$

c. $\frac{140}{13}$ ✓

d. $\frac{153}{13}$

$$d = \frac{|5(9) + 12(7) + 11|}{\sqrt{25 + 144}} = \frac{|140|}{13} = \frac{140}{13}.$$

- 36 Find the distance from the point $(3, -8)$ to the line $12x + 35y - 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $-\frac{253}{37}$

b. $\frac{253}{37}$ ✓

c. $-\frac{253}{47}$

d. $\frac{253}{47}$

$$d = \frac{|12(3) + 35(-8) - 9|}{\sqrt{144 + 1225}} = \frac{|-253|}{37} = \frac{253}{37}.$$

37 Find the distance from the point $(-2, -2)$ to the line $7x - 24y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $-\frac{19}{17}$

b. $\frac{44}{25}$

c. $\frac{69}{25}$

d. $\frac{19}{25}$ ✓

$$d = \frac{|7(-2) - 24(-2) - 15|}{\sqrt{49 + 576}} = \frac{|19|}{25} = \frac{19}{25}.$$

38 Find the distance from the point $(8, 0)$ to the line $12x - 35y - 8 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{162}{37}$

b. $\frac{125}{37}$

c. $\frac{88}{37}$ ✓

d. $-\frac{88}{23}$

$$d = \frac{|12(8) - 35(0) - 8|}{\sqrt{144 + 1225}} = \frac{|88|}{37} = \frac{88}{37}.$$

39 Find the distance from the point $(-3, 5)$ to the line $12x + 35y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{161}{37}$

b. $\frac{124}{47}$

c. $\frac{198}{37}$

d. $\frac{124}{37}$ ✓

$$d = \frac{|12(-3) + 35(5) - 15|}{\sqrt{144 + 1225}} = \frac{|124|}{37} = \frac{124}{37}.$$

- 40 Find the distance from the point $(-8, 0)$ to the line $9x + 40y + 6 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $-\frac{66}{49}$

b. $\frac{66}{49}$

c. $-\frac{66}{41}$

d. $\frac{66}{41}$ ✓

$$d = \frac{|9(-8) + 40(0) + 6|}{\sqrt{81 + 1600}} = \frac{|-66|}{41} = \frac{66}{41}.$$

- 41 Find the distance from the point $(-7, -8)$ to the line $12x - 35y + 10 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{280}{37}$

b. $\frac{243}{37}$

c. $\frac{206}{37}$ ✓

d. $-\frac{206}{23}$

$$d = \frac{|12(-7) - 35(-8) + 10|}{\sqrt{144 + 1225}} = \frac{|206|}{37} = \frac{206}{37}.$$

- 42 Find the distance from the point $(-5, 5)$ to the line $8x + 15y - 13 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{39}{17}$

b. $\frac{22}{23}$

c. $\frac{22}{17}$ ✓

d. $\frac{56}{17}$

$$d = \frac{|8(-5) + 15(5) - 13|}{\sqrt{64 + 225}} = \frac{|22|}{17} = \frac{22}{17}.$$

- 43 Find the distance from the point $(7, 1)$ to the line $3x + 4y + 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{44}{5}$

b. $\frac{39}{5}$

c. $\frac{34}{7}$

d. $\frac{34}{5}$ ✓

$$d = \frac{|3(7) + 4(1) + 9|}{\sqrt{9 + 16}} = \frac{|34|}{5} = \frac{34}{5}.$$

- 44 Find the distance from the point $(2, -3)$ to the line $20x + 21y + 7 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

- a. $\frac{16}{41}$
- b. $-\frac{16}{41}$
- c. $-\frac{16}{29}$
- d. $\frac{16}{29}$ ✓

$$d = \frac{|20(2) + 21(-3) + 7|}{\sqrt{400 + 441}} = \frac{|-16|}{29} = \frac{16}{29}.$$

- 45 Find the distance from the point $(2, -2)$ to the line $5x - 12y + 12 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

- a. $\frac{46}{13}$ ✓
- b. $\frac{59}{13}$
- c. $\frac{72}{13}$
- d. $-\frac{46}{7}$

$$d = \frac{|5(2) - 12(-2) + 12|}{\sqrt{25 + 144}} = \frac{|46|}{13} = \frac{46}{13}.$$

Multiple choice — hard

- 46 Find the y-intercept of the perpendicular bisector of the segment connecting $(-2, -1)$ and $(4, -6)$.

- a. $-\frac{47}{10}$ ✓
- b. $-\frac{8}{3}$
- c. $-\frac{37}{10}$
- d. $\frac{47}{10}$

Midpoint: $(1, -7/2)$. Slope of the segment: $-5/6$, so the perpendicular slope is $6/5$. Using $y + 7/2 = 6/5(x - 1)$, the y-intercept is $-\frac{47}{10}$.

- 47 Find the y-intercept of the perpendicular bisector of the segment connecting $(-5, -6)$ and $(0, 0)$.

- a. $\frac{61}{12}$
- b. 0
- c. $-\frac{61}{12}$ ✓
- d. $-\frac{49}{12}$

Midpoint: $(-5/2, -3)$. Slope of the segment: $6/5$, so the perpendicular slope is $-5/6$. Using $y + 3 = -5/6(x + 5/2)$, the y-intercept is $-\frac{61}{12}$.

- 48 Find the y-intercept of the perpendicular bisector of the segment connecting $(8, 7)$ and $(6, -5)$.

- a. $-\frac{13}{6}$
- b. -41
- c. $\frac{13}{6}$ ✓
- d. $\frac{19}{6}$

Midpoint: $(7, 1)$. Slope of the segment: 6, so the perpendicular slope is $-1/6$. Using $y - 1 = -1/6(x - 7)$, the y-intercept is $\frac{13}{6}$.

- 49 Find the y-intercept of the perpendicular bisector of the segment connecting $(2, -3)$ and $(8, 7)$.

- a. -5
- b. 5 ✓
- c. 6
- d. $-\frac{19}{3}$

Midpoint: $(5, 2)$. Slope of the segment: $5/3$, so the perpendicular slope is $-3/5$. Using $y - 2 = -3/5(x - 5)$, the y-intercept is 5.

- 50 Find the y-intercept of the perpendicular bisector of the segment connecting $(-6, 0)$ and $(6, -3)$.

- a. $-\frac{3}{2}$ ✓
- b. $\frac{1}{2}$
- c. $\frac{3}{2}$
- d. $-\frac{1}{2}$

Midpoint: $(0, -\frac{3}{2})$. Slope of the segment: $-\frac{1}{4}$, so the perpendicular slope is 4. Using $y + \frac{3}{2} = 4(x)$, the y-intercept is $-\frac{3}{2}$.

- 51 Find the sum of the x-coordinates of the intersection points of $y = x^2 - 12$ and $y = -4x$.

- a. -12
- b. -4 ✓
- c. -2
- d. -6

Setting the equations equal: $x^2 + 4x - 12 = 0$. By Vieta's formulas, the sum of the roots is $-(4) = -4$.

- 52 Find the sum of the x-coordinates of the intersection points of $y = x^2 - 4$ and $y = -4x + 1$.

- a. -2
- b. -5
- c. -6
- d. -4 ✓

Setting the equations equal: $x^2 + 4x - 5 = 0$. By Vieta's formulas, the sum of the roots is $-(4) = -4$.

53 Find the sum of the x -coordinates of the intersection points of $y = x^2 + 2x + 21$ and $y = -6x + 9$.

a. -10

b. -6

c. -8 ✓

d. 12

Setting the equations equal: $x^2 + 8x + 12 = 0$. By Vieta's formulas, the sum of the roots is $-(8) = -8$.

54 Find the sum of the x -coordinates of the intersection points of $y = x^2 + 4x - 11$ and $y = -x - 5$.

a. -3

b. -5 ✓

c. -6

d. -7

Setting the equations equal: $x^2 + 5x - 6 = 0$. By Vieta's formulas, the sum of the roots is $-(5) = -5$.

55 Find the sum of the x -coordinates of the intersection points of $y = x^2 + 15x + 28$ and $y = 4x - 2$.

a. -9

b. -13

c. 30

d. -11 ✓

Setting the equations equal: $x^2 + 11x + 30 = 0$. By Vieta's formulas, the sum of the roots is $-(11) = -11$.

56 A parabola $y = ax^2 + bx + c$ passes through $(0, 5)$, $(1, 8)$, and $(-1, -4)$. Find a .

a. -3 ✓

b. 6

c. -2

d. -4

From $(0, 5)$: $c = 5$. From $(1, 8)$: $a + b + c = 8$, so $a + b = 3$. From $(-1, -4)$: $a - b + c = -4$, so $a - b = -9$. Adding: $2a = -6$, so $a = -3$.

57 A parabola $y = ax^2 + bx + c$ passes through $(0, 5)$, $(1, 2)$, and $(-1, 2)$. Find a .

a. -4

b. -2

c. 0

d. -3 ✓

From $(0, 5)$: $c = 5$. From $(1, 2)$: $a + b + c = 2$, so $a + b = -3$. From $(-1, 2)$: $a - b + c = 2$, so $a - b = -3$. Adding: $2a = -6$, so $a = -3$.

58 A parabola $y = ax^2 + bx + c$ passes through $(0, -3)$, $(1, -1)$, and $(-1, -11)$. Find a .

a. -4

b. -3 ✓

c. -2

d. 5

From $(0, -3)$: $c = -3$. From $(1, -1)$: $a + b + c = -1$, so $a + b = 2$. From $(-1, -11)$: $a - b + c = -11$, so $a - b = -8$. Adding: $2a = -6$, so $a = -3$.

59 A parabola $y = ax^2 + bx + c$ passes through $(0, 7)$, $(1, 0)$, and $(-1, 12)$. Find a .

a. -6

b. -1 ✓

c. 0

d. -2

From $(0, 7)$: $c = 7$. From $(1, 0)$: $a + b + c = 0$, so $a + b = -7$. From $(-1, 12)$: $a - b + c = 12$, so $a - b = 5$. Adding: $2a = -2$, so $a = -1$.

60 A parabola $y = ax^2 + bx + c$ passes through $(0, -3)$, $(1, -2)$, and $(-1, -2)$. Find a .

a. 1 ✓

b. 2

c. 0

d. 3

From $(0, -3)$: $c = -3$. From $(1, -2)$: $a + b + c = -2$, so $a + b = 1$. From $(-1, -2)$: $a - b + c = -2$, so $a - b = 1$. Adding: $2a = 2$, so $a = 1$.